

Zixun Huang

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EDUCATION

- **Wharton School, University of Pennsylvania (Admitted)** Aug. 2026 - Jun. 2031
 - *Ph.D. in Statistics*
- **Peking University, School of Mathematical Sciences** Sept. 2022 – Jun. 2026
 - *B.S. in Statistics*, GPA: 3.7/4.0
 - Member of Elite Undergraduate Program for Applied Mathematics (**Top 10%**)
 - Honors and Awards: Outstanding Student Award (2023 & 2024), First Prize in Chinese Mathematics Contest (2024, top 1%), Yau Contest Groups Prize (2024), Gold Prize in Chinese Mathematical Olympiad (2021), Silver Prize in Chinese Mathematical Olympiad (2020)
- **University of California, Berkeley** Jan. 2025 – Aug. 2025
 - *Visiting Student*, GPA: 4.0/4.0

PUBLICATIONS

- Binghui Li*, Fengling Chen*, **Zixun Huang***, Lean Wang*, Lei Wu. *Functional Scaling Laws in Kernel Regression: Loss Dynamics and Learning Rate Schedules*. **NeurIPS 2025 (Spotlight)**. [Paper]
- **Zixun Huang***, Jiayi Sheng*, Zeyu Zheng. *OBLR-PO: A Theoretical Framework for Stable Reinforcement Learning*. Submitted to **AISTATS 2026 (Reviewer scores: 6, 5, 5 (max 7))**. [Paper]
- Yuhao Sun, Zekun Wu, **Zixun Huang**, Peijie Zhou. *TracingFlow: A Simulation-Free Trajectory Inference Framework Based on Second-Order Dynamics*. Submitted to **ICML 2026**.

RESEARCH EXPERIENCE

Core Research Focus: Modeling complex systems using mathematical and statistical methods, and analyzing optimal strategies for training and inference in large language models.

- **Functional Scaling Laws for Learning Rate Schedules** May. 2024 – Nov. 2025
 - Advisor: *Prof. Lei Wu*
 - Proposed a novel Functional Scaling Law (FSL) by modeling SGD as an intrinsic-time SDE
 - Analyzed learning-rate schedules under data and compute constraints, revealing efficiency trade-offs
 - Validated FSL through kernel regression and LLM pre-training, producing optimal schedules
- **OBLR-PO: Theory for Stable Reinforcement Learning** Feb. 2025 – Oct. 2025
 - Advisor: *Prof. Zeyu Zheng*
 - Established theoretical guarantees for policy optimization, including loss bounds and convergence
 - Introduced adaptive learning rate and gradient-weighted baselines for principled variance reduction
 - Implemented OBLR-PO and demonstrated superior performance compared with existing algorithms
- **Second Order Flow Matching** Sept. 2025 – Feb. 2026
 - Advisor: *Prof. Peijie Zhou*
 - Proposed TracingFlow, a simulation-free second-order flow matching framework.
 - Formulated Dynamical Optimal Acceleration Transport (DOAT) via acceleration field regression.
 - Demonstrated improved trajectory inference and distribution reconstruction on single-cell datasets.

SKILLS AND OTHERS

- **TOEFL iBT** Total 101: 28(R), 23(L), 23(S), 27(W)
- **Quantitative Skills** GRE General: V160, Q170; GRE Math: 970 (95th percentile)
- **Programming** Python(NumPy, Pandas, Scikit-learn), MATLAB, LaTeX, PyTorch, C++